

Mathematics A

Unit 1 • Chapter OL-T09 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

4 classified items • 16 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T09 • Expertise Q16+ • 4 items • 16 marks

Chapter	Items	Marks	Starts
01. OL-T09 Rounding, Estimation & Bounds	4	16	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Rounding, Estimation & Bounds

Unit 1 • Expertise • 16 marks • 4 item(s)

June 2025 | 4MA1-2H Q22 | 4 marks

Geometry, rate or practical decision bounds

November 2024 | 4MA1-1H Q21 | 3 marks

Algebraic expression upper or lower bound

May-Nov 2020 | 1H Q17 | 4 marks

Mixed stated accuracies

January 2023 | 1H Q19 | 5 marks

Round to decimal places or significant figures

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q22

4 marks

22 The diagram shows a square inside rectangle $ABCD$

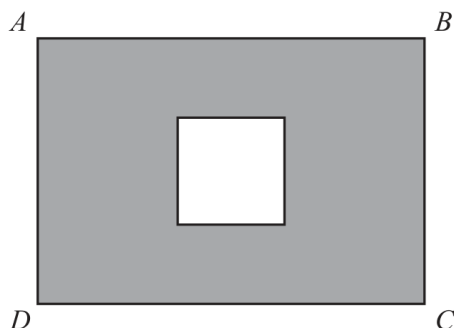


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X \text{ cm}^2$

$AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2 \text{ cm}$ correct to 2 significant figures

side of square = 4.1 cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

WORKING SPACE

4MA1-2H Q22

4 marks

22 The diagram shows a square inside rectangle $ABCD$

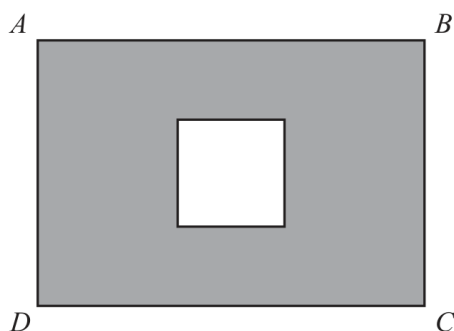


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By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

Bounds for a shaded area**4 marks****Input bounds**

$$11.25 \leq AB < 11.75, \quad 9.15 \leq BC < 9.25, \quad 4.05 \leq s < 4.15.$$

Upper shaded area

$$X_U = 11.75(9.25) - 4.05^2 = 92.285.$$

Lower shaded area

$$X_L = 11.25(9.15) - 4.15^2 = 85.715.$$

Choose the safe accuracy

$$85.715 \leq X < 92.285; \text{ both bounds round to 90 to 1 significant figure.}$$

Final answer

$$X = 90 \text{ cm}^2 \text{ (1 s.f.)}$$

4MA1-1H Q21 M01

3 marks

$$21 \quad T = \frac{x^2 + y^2}{w}$$

 $x = 28.4$ correct to 1 decimal place. $y = 17$ correct to 2 significant figures. $w = 90$ correct to the nearest 5Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.

Show your working clearly.

.....

WORKING SPACE

Use upper and lower bounds

3 marks

M01 – Use upper and lower bounds**Write the bounds**

$$UB(x) = 28.45, UB(y) = 17.5, LB(w) = 87.5.$$

Substitute

$$UB(T) = \frac{28.45^2 + 17.5^2}{87.5} = 12.7503\dots$$

Round

$$UB(T) = 12.8 \text{ (3 s.f.)}.$$

Final answer

12.8

- 17** A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3}) \text{ m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

..... kg/m^3

(Total for Question 17 is 4 marks)

Answer reference | May-Nov 2020 | Question 17

Family: Recover single-measurement bounds • Variant: Mixed stated accuracies

17	5025 or 5.025 or 4975 or 4.975		4	B1 Accept 5024.9 for 5025 or 5.0249 for 5.025
	1.845×10^{-3} oe or 1.835×10^{-3} oe			B1 Accept 1.8449×10^{-3} for 1.845×10^{-3}
	$\frac{5.025}{1.835 \times 10^{-3}} (= 2738.4...) \text{ oe}$			M1 for correct substitution into $\frac{m_{UB}}{v_{LB}}$ where $5 < m_{UB} \leq 5.025$ and $1.835 \times 10^{-3} \leq v_{LB} < 1.84 \times 10^{-3}$
		2738.4		A1 dep on correct working
				Total 4 marks

Worked solution

May-Nov 2020 | Question 17 | 4 marks

Family: Recover single-measurement bounds • Variant: Mixed stated accuracies

Method

Density is

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

To get the upper bound for density, use the upper bound for mass and the lower bound for volume.

The mass is 5 kg correct to the nearest 50 g.

$$50 \text{ g} = 0.05 \text{ kg}$$

Half of this is 0.025 kg, so

$$\text{upper bound for mass} = 5 + 0.025 = 5.025$$

The volume is $1.84 \times 10^{-3} = 0.00184 \text{ m}^3$, correct to 3 significant figures.

The lower bound is

$$0.00184 - 0.000005 = 0.001835$$

So the upper bound for the density is

$$(5.025) / (0.001835) = 2738.419 \dots$$

Correct to 1 decimal place,

$$2738.4$$

Answer: 2738.4 kg/m^3

19 The acceleration, a , of an object is given by

$$a = \frac{v - u}{t}$$

where

$v = 45.23$ correct to 2 decimal places

$u = 5.12$ correct to 2 decimal places

$t = 8.5$ correct to 2 significant figures

By considering bounds, work out the value of a to a suitable degree of accuracy.
Show your working clearly and give a reason for your answer.

$a = \dots\dots\dots$

(Total for Question 19 is 5 marks)

Answer reference | January 2023 | Question 19

Family: Round to a stated accuracy • Variant: Round to decimal places or significant figures

19	45.225 or 45.235 or 5.115 or 5.125 or 8.45 or 8.55		5	B2 for all 6 correct (B1 for 4 or 5 correct) Accept $45.234\dot{9}$ for 45.235 $5.124\dot{9}$ for 5.125 $8.54\dot{9}$ for 8.55
	$\frac{45.235 - 5.115}{8.45} (= 4.7479\dots)$			M1 for correct substitution into the <i>UB</i> $a = \frac{v-u}{t}$ where $45.23 < v_{(UB)} \leq 45.235$ $5.115 \leq u_{(LB)} < 5.12$ $8.45 \leq t_{(LB)} < 8.5$
	$\frac{45.225 - 5.125}{8.55} (= 4.6900\dots)$			M1 for correct substitution into the <i>LB</i> $a = \frac{v-u}{t}$ where $45.225 \leq v_{(LB)} < 45.23$ $5.12 < u_{(UB)} \leq 5.125$ $8.5 < t_{(UB)} \leq 8.55$
	Working required	4.7 and correct reason		A1 dep on M2 4.7 and both answers round to 4.7 oe e.g. 1 dp or 2 sf
Total 5 marks				

Worked solution

January 2023 | Question 19 | 5 marks

Family: Round to a stated accuracy • Variant: Round to decimal places or significant figures

Official final mark-scheme pages are embedded immediately after this question.